Day 01 Short Question

Question 1: Define a vector space R. State the two closure properties.

Solution: A set V is a vector space over R if it is equipped with vector addition and scalar multiplication satisfy the vector space.

Closure properties:

Addition closure: $v(u+v) = uv + vv$

Scalar closure: $c(uv) = u(cv)$

Question 2: What is a vector in R^n?

Solution: A vector in R^n is an ordered tuple of n real number

$$X = [x_1, x_2, \ldots x_n]^T$$

3) Define a subspace. Three conditions.

Solution: A subspace W of a vector V is a subset

$W \subseteq V$.

operations are

- 0 in W
- If u, v in W then u+v in W
- If a in R an v in W then a*v in W.

Question 4: Linear combination of vectors (general form).

Solution:

A linear combination of vector $v_1, v_2 \cdots v_k$ is

$$a_1 v_1 + a_2 v_2 - - - - a_k v_k$$

Where $a_1 - - - a_k$ in R.

Question 5: Span defination + geometric meaning.

Solution: The span of $\{v_1 \cdots v_k\}$ is the set of all linear combinations of them:

$$\text{Span}(v_1 \cdots v_k) = \{a_1 * v_1 + \cdots a_k * v_k : a_1 \cdots a_k \text{ in } R\}$$

Qustion 6: Linear indepence + Condition.

Solution: vector $\{v_1, v_2 \cdots v_k\}$ are Linearly independt if the only way to get 0 is the trivial coefficient

Question 7: Linear Dependence + Why bad in ML

Solution: It means at least one vector is a linear combination of other. there exist not all zero scalar $a_1 v_1 + \cdots a_k v_k$ and $a_i \neq 0$.

ML problem: It creats redednat features, making parameter unstable, causing numerical issue.

Question 8: Basis defination + two properties.

Solution: A Basis is a set of vectors.

1) They span the space.

2) They are linenly independent

<u>Qustion 9:</u> Dimension meaning + relation to basis

<u>Solution:</u> The dimension of a vector space is the no of vector in any basis of that space.

So if basis d vectors, then $\dim(V) = d$.

<u>Question 10:</u> Matrix Rank meaning + relation to independec.

<u>Solution:</u> Matrix Rank is a no of linenly independent columns.

So, $\text{rank}(A) = r$ means there are r independent vectors among its columns

Column are linenly independt iff $\text{rank}(A) = \text{no_of_cols}$

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